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GeoSosEkon: An Multimodal Integrated Spatial-Temporal Analytics System for Modeling Provincial Poverty Dynamics in Indonesia Using UMAP, PCA, GMM, Ensemble Forecasting, SHAP, Moran's I Analysis, LISA, Fixed Effect Panel Data Regression and Sentiment Analysis using X with RoBERTa

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Abstract. This study presents GeoSosEkon, a multimodal integrated spatial-temporal analytics system for modeling provincial poverty dynamics in Indonesia using socioeconomic panel data from Badan Pusat Statistik (BPS) covering 38 provinces from 2021 to 2024 and 521 poverty-related tweets scraped from X (formerly Twitter) spanning 2021 to 2026. The system operates through five parallel analytical modules: dimensionality reduction and probabilistic clustering via comparative UMAP and PCA preprocessing for Gaussian Mixture Models, which confirmed UMAP's superiority with an optimal Silhouette Score of 0.822 at k=2; ensemble forecasting using Random Forest, XGBoost, and LightGBM with weighted averaging yielding a MAPE of 4.57%, interpreted through adaptive SHAP analysis identifying prior poverty level and human development attainment as dominant predictors; spatial autocorrelation analysis via Global Moran's I consistently exceeding 0.57 across all years and LISA classification revealing a persistent High-High poverty hotspot across five to six Papua provinces throughout the observation period; Two-Way Fixed Effect panel regression establishing mean years of schooling as the most robust within-province causal determinant of poverty reduction with a coefficient of -1.225 (p = 0.000) after controlling for province and time fixed effects; and IndoRoBERTa-based sentiment analysis revealing that public discourse on poverty is predominantly neutral (65.5%) with a growing negative undercurrent reaching 45.0% in 2026. Collectively, the results confirm that Indonesian provincial poverty is geographically self-reinforcing, educationally determined at the within-province level, and spatially concentrated in the eastern archipelago, providing an empirical foundation for geographically differentiated and education-centered poverty alleviation strategies.

Keywords: provincial poverty, spatial autocorrelation, Gaussian Mixture Model, fixed effect panel regression, sentiment analysis

Chapter I Introduction

Poverty remains a deeply entrenched structural challenge in Indonesia, a nation of over 270 million people spread across 38 provinces with pronounced socioeconomic heterogeneity. Recent data indicate that provincial poverty rates range from 4.53% to 26.80%, while human capital indicators such as ICT skills vary from 29.82% to 92.36%, reflecting long-standing structural disparities

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in the Indonesian developmental trajectory [1]. Despite consistent poverty reduction trends observed over the past decade, significant inequality persists among provinces, particularly in the eastern regions of Indonesia [2]. These disparities are not merely economic in nature, they are deeply spatial. Spatial autocorrelation analysis has shown that the distribution pattern of poverty across Indonesian regencies and cities is clustered rather than random, forming what researchers term poverty pockets that do not correspond to government administrative boundaries, thus necessitating integrative rather than uniform regional policy approaches [3]. This geographic concentration implies that poverty alleviation strategies must account for cross-provincial interdependencies and spillover effects that standard non-spatial analytical frameworks cannot capture.

To address these analytical gaps, machine learning methods have been increasingly adopted for poverty modeling. Studies applying machine learning techniques to forecast poverty at the district and city level using BPS socioeconomic data have demonstrated the viability of data-driven approaches to poverty prediction in Indonesia [4]. Among ensemble methods, Random Forest and Extreme Gradient Boosting (XGBoost) have been applied specifically to forecast poverty as supervised ensemble learning classification approaches, owing to their robustness and interpretability in socioeconomic contexts [5]. The choice of dimensionality reduction technique prior to clustering also critically affects analytical outcomes. Comparative studies across 240 synthetic datasets have shown that UMAP outperforms PCA in clustering quality, with UMAP achieving a mean Silhouette coefficient of 0.73 compared to 0.46 for PCA, while PCA shows a negative association between performance and the number of clusters to recover [6]. Nevertheless, PCA retains advantages in interpretability and stability for small sample sizes, motivating the need for empirical comparison rather than a priori selection. Alongside these clustering approaches, causal inference remains essential. Two-way fixed-effects panel modeling is widely regarded as the gold standard for leveraging longitudinal data structure to infer causality from within-unit effects, effectively controlling for time-invariant heterogeneity across observational units [7].

Beyond quantitative tabular analysis, public discourse on poverty increasingly manifests on social media platforms, offering a complementary layer of qualitative insight into societal perceptions of poverty dynamics. RoBERTa-based models have demonstrated strong performance for multi-class sentiment and emotion classification on Indonesian Twitter data, establishing IndoRoBERTa as an effective architecture for Indonesian social media text analysis [8]. Integrating such sentiment signals alongside spatial, econometric, and ensemble machine learning components represents the core methodological contribution of this project. GeoSosEkon therefore synthesizes UMAP and PCA comparison for GMM clustering, ensemble forecasting with Random Forest, XGBoost, and LightGBM, adaptive SHAP for interpretability, Moran's I and LISA for spatial autocorrelation, two-way fixed-effect panel regression for causal inference, and RoBERTa-based sentiment analysis on poverty-related posts scraped from X (formerly Twitter) into a unified multimodal spatial-temporal analytics system for modeling provincial poverty dynamics in Indonesia from 2021 to 2024.

Chapter 2 Research Methodology

A. Data Explanation

This study utilizes two complementary datasets reflecting the multimodal nature of the analytical framework. The primary dataset is tabular socioeconomic data sourced from Badan Pusat Statistik (BPS), comprising 151

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observations across 38 Indonesian provinces from 2021 to 2024, with 2021 containing 37 observations due to one missing provincial record. Ten variables are selected after preprocessing: School Participation Rate for age brackets 13–15, 16–18, and 19–24 (APS); averaged Open Unemployment Rate (TPT) and Labor Force Participation Rate (TPAK) from February and August measurements; averaged Poverty Percentage and Poverty Line from March and September measurements; Human Development Index (HDI); Mean Years of Schooling (RLS); and Expected Years of Schooling (HLS). The dependent variable is the Percentage of Poor Population, ranging from 4.00% to 32.97% across the dataset. The secondary dataset consists of 521 Indonesian-language tweets scraped from X (formerly Twitter) using the keyword "persentase penduduk miskin," covering 2021–2026 with 94, 100, 100, 87, 100, and 40 tweets per year respectively. Since the location field contains no non-null values across all records, this dataset serves exclusively as a national-level temporal sentiment signal rather than a province-level source.

B. Research Variables

Table 1. Research Variables and Descriptions

Variable Name	Explanation
APS 13-15	Angka Partisipasi Sekolah 13–15 (School participation rate for ages 13–15, showing the proportion of adolescents enrolled in education)
APS 16-18	Angka Partisipasi Sekolah 16–18 (School participation rate for ages 16–18, indicating the percentage of older teens attending school)
APS 19-24	Angka Partisipasi Sekolah 19–24 (School participation rate for ages 19–24, reflecting youth participation in higher education or vocational training)
TPT - Februari	Tingkat Pengangguran Terbuka Februari (Open unemployment rate in February, representing the share of the labor force without work but actively seeking employment)
TPT - Agustus	Tingkat Pengangguran Terbuka Agustus (Open unemployment rate in August, capturing seasonal or mid-year unemployment trends)
TPAK - Februari	Tingkat Partisipasi Angkatan Kerja Februari (Labor force participation rate in February, showing the proportion of the working-age population engaged or seeking work)
TPAK - Agustus	Tingkat Partisipasi Angkatan Kerja Agustus (Labor force participation rate in August, reflecting workforce activity during mid-year)
Persentase Penduduk Miskin - Maret	Poverty rate in March, indicating the percentage of people living below the poverty line
Persentase Penduduk Miskin - September	Poverty rate in September, showing seasonal variation or mid-year poverty levels

Indeks Pembangunan Manusia	Human Development Index, summarizing achievements in health, education, and income
Harapan Lama Sekolah (Tahun)	Expected years of schooling, estimating the total years a child entering school is expected to spend in education
.....	.....

C. Research Workflow

The GeoSosEkon pipeline operates through two parallel ingestion streams converging into five independent analytical modules, as illustrated in Figure 1. The BPS tabular stream undergoes preprocessing - including dual-period averaging for TPT, TPAK, poverty rate, and poverty line, followed by standardization - before entering four analytical modules: dimensionality reduction and GMM clustering (Module A), ensemble forecasting with SHAP interpretability (Module B), spatial autocorrelation analysis (Module C), and two-way fixed effect panel regression (Module D). Concurrently, the Twitter stream undergoes text cleaning, stopword removal, and tokenization before entering RoBERTa-based sentiment analysis (Module E). All module outputs are subsequently synthesized in a cross-module integration stage and rendered into an interactive web dashboard for policy-oriented exploration.

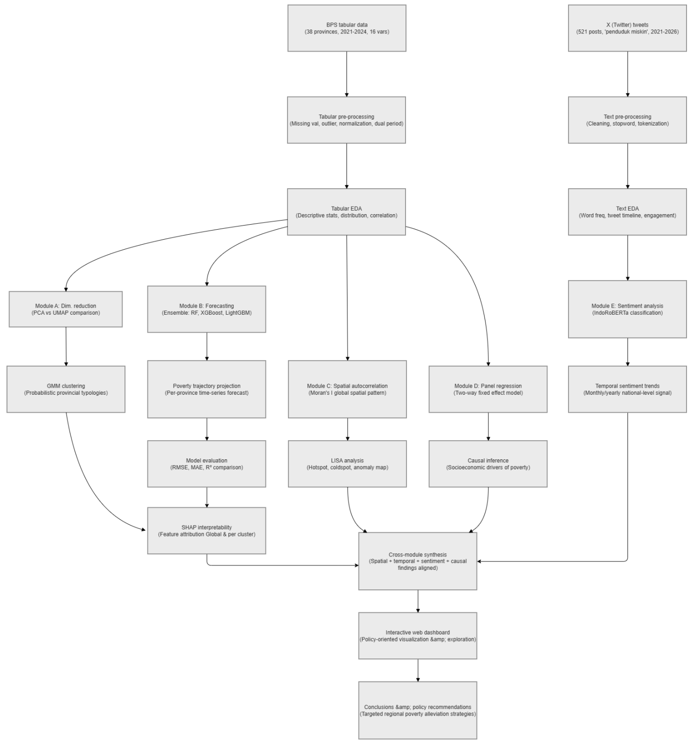

Figure 1. GeoSosEkon Research Workflow

D. Dimensionality Reduction and GMM Clustering

PCA and UMAP are applied as competing dimensionality reduction strategies prior to clustering, both reducing the 10 standardized socioeconomic indicators into a three-dimensional embedding space. PCA transforms the feature matrix into orthogonal principal components by maximizing explained variance via $z = Xw$, where w is the eigenvector of the covariance matrix. UMAP preserves local and global manifold structure by optimizing cross-entropy between high-dimensional and low-dimensional probability distributions [6]. Cluster quality across $k = 2$ to 10 is evaluated using Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Score before fitting Gaussian Mixture Models (GMM), which model the data as a weighted mixture of Gaussian distributions [9]:

$$L(\mu, \Sigma, \pi) = \prod_{i=1}^{N} \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}(x_i \mid \mu_k, \Sigma_k)$$

where π_k, μ_k, and Σ_k denote the mixing weight, mean, and covariance of cluster k, with parameters estimated via the

Expectation-Maximization algorithm enabling soft probabilistic provincial assignment.

E. Ensemble Forecasting and SHAP Interpretability

Ensemble forecasting is implemented using Random Forest (RF), XGBoost, and LightGBM to predict provincial poverty rates from eight socioeconomic features, with predictions combined via weighted averaging where each model's weight is derived from its inverse cross-validation RMSE [10]. RF constructs multiple bootstrap decision trees; XGBoost and LightGBM iteratively minimize residual errors through gradient boosting, with LightGBM applying histogram-based leaf-wise tree growth for computational efficiency. Forecasting targets poverty rate projections for 2025–2026. Model interpretation uses SHAP (SHapley Additive Explanations), which quantifies each feature's marginal contribution [11]:

$$\varphi_j(f, x) = \sum_{S \subseteq F \setminus \{j\}} \left[|S|!(|F|-|S|-1)! \,/\, |F|! \right] \cdot [f(S \cup \{j\}) - f(S)]$$

ensuring consistency and local accuracy in feature attribution, with results visualized through beeswarm and waterfall plots to identify the most influential socioeconomic indicators globally and per provincial cluster.

F. Spatial Autocorrelation

Spatial autocorrelation analysis examines geographic clustering of poverty across Indonesian provinces using Global Moran's I and Local Indicators of Spatial Association (LISA), with a k-nearest neighbor spatial weights matrix (k=5) constructed from provincial centroid coordinates. Significance is tested using 999 random permutations, where positive Moran's I indicates spatial clustering, values near zero indicate randomness, and negative values indicate dispersion [12]:

$$I = (n \,/\, \Sigma_i \Sigma_j w_{ij}) \cdot (\Sigma_i \Sigma_j w_{ij}(x_i - \bar{x})(x_j - \bar{x})) \,/\, (\Sigma_i (x_i - \bar{x})^2)$$

LISA then decomposes this global statistic into local province-level contributions, classifying each province into High-High (poverty hotspot), Low-Low (coldspot), High-Low, or Low-High spatial regimes to identify poverty concentration and regional anomalies.

G. Two-Way Fixed Effect Panel Regression

Two-Way Fixed Effect (TWFE) panel regression estimates the causal relationship between socioeconomic indicators and provincial poverty rates while controlling for unobserved heterogeneity across provinces and time periods [13]:

$$y_{it} = \alpha_i + \lambda_t + \beta X_{it} + \varepsilon_{it}$$

where α_i captures province-specific fixed effects, λ_t captures year fixed effects, and X_{it} represents time-varying socioeconomic covariates. Standard errors are clustered at the province level to address serial correlation, and model selection between Fixed Effects and Random Effects is validated using the Hausman specification test, with robustness checks against Pooled OLS and entity-only Fixed Effects.

H. Sentiment Analysis with IndoRoBERTa

Sentiment analysis is applied to 521 poverty-related tweets using the pre-trained Indonesian RoBERTa model (w11wo/indonesian-roberta-base-sentiment-classifier), which improves upon BERT through dynamic masking, larger batch training, and removal of the Next Sentence Prediction objective [8]. Tweet text is tokenized into subword units and processed through a classification head producing sentiment probabilities via:

$$P(c \mid x) = e^{z_c} \,/\, \Sigma_{c'} e^{z_{c'}}$$

where z_c is the logit score for class c. Inference is performed on GPU in batches of 32, assigning sentiment labels (positive, neutral, negative) and confidence scores to each tweet. Results are aggregated by month and year to produce a temporal trend visualization of public discourse on poverty from 2021 to 2026.

Chapter 3 Results Analysis

A. Exploratory Data Analysis

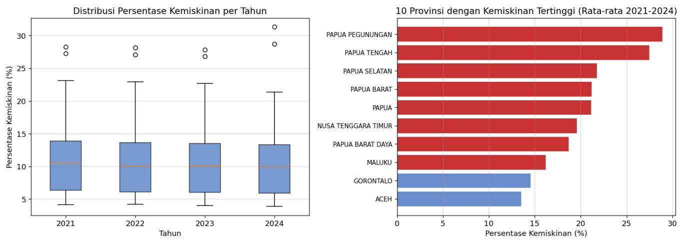

Figure 2A. Historical Poverty Analysis

Figure 2A (left) presents the annual distribution of provincial poverty rates from 2021 to 2024 through box plots. The median poverty rate remained consistently stable across all four years at approximately 10%, suggesting that while the central tendency of provincial poverty showed minimal change over the observation period, the interquartile range remained notably wide, spanning roughly 7% to 14%, indicating persistent structural disparity across provinces. Outlier provinces are visible in every year, with values reaching above 27% to 31%, reflecting the presence of extreme cases that pull the upper tail of the distribution significantly beyond the bulk of observations. The gradual narrowing of the upper whisker from 2021 to 2024 hints at modest improvement in the most affected provinces, though the overall distributional shape remained largely unchanged, implying that national-level poverty reduction efforts have yet to substantially compress inter-provincial inequality.

Figure 2A (right) identifies the ten provinces with the highest average poverty rates over 2021 to 2024. The ranking is dominated by provinces in the Papua region, with Papua Pegunungan recording the highest average at 28.88%, followed by Papua Tengah at 27.45%, Papua Selatan at 21.79%, Papua Barat at 21.21%, and Papua at 21.13%. Nusa Tenggara Timur (19.57%), Papua Barat Daya (18.70%), and Maluku (16.18%) complete the eastern Indonesia cluster, all highlighted in red, reinforcing the strong geographic concentration of poverty in the eastern archipelago. Gorontalo (14.53%) and Aceh (13.54%), shown in blue, are the only provinces outside the Papua-Maluku-NTT corridor to appear in the top ten, suggesting that while eastern Indonesia constitutes the primary poverty hotspot, pockets of high poverty also persist in Sulawesi and Sumatra. This spatial concentration provides the primary empirical motivation for the spatial autocorrelation analysis conducted in subsequent sections.

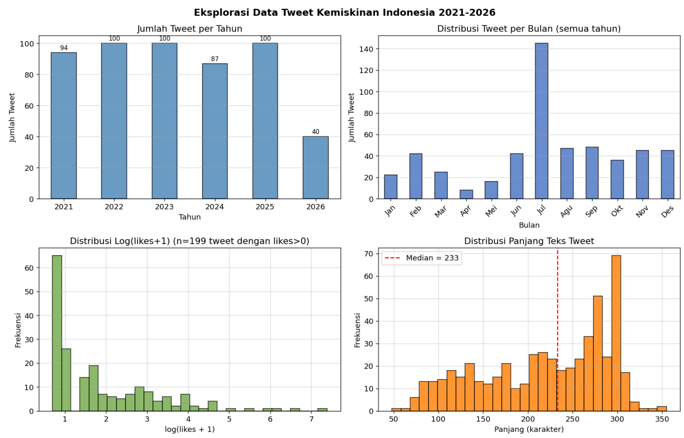

Figure 2B. Exploratory Analysis of Poverty-Related Tweet Data from Indonesia (2021-2026)

Figure 2B presents the exploratory analysis of the 521 poverty-related tweets scraped from X (formerly Twitter) using the keyword "persentase penduduk miskin" covering the period from 2021 to 2026. The top left panel displays the annual tweet count distribution, showing relatively stable scraping volumes of 94 tweets in 2021, 100 in each of 2022, 2023, and 2025, 87 in 2024, and 40 in 2026, with the lower count in 2026 attributable to the incomplete scraping window rather than a genuine reduction in public discourse activity. The top right panel reveals the monthly distribution of tweets aggregated across all years, showing a highly pronounced spike in July with approximately 143 tweets, more than three times the volume observed in the lowest month of April which recorded only around 9 tweets. This July concentration is consistent with the BPS data release cycle, as the agency

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typically publishes its March poverty survey results in mid-year, triggering a surge of public commentary and media reporting on poverty statistics that is captured by the keyword-based scraping approach. Secondary peaks are visible in August, September, November, and December, months that also correspond to periods of policy announcements, national budget deliberations, and the lead-up to the September BPS poverty survey measurement period.

The bottom two panels provide insight into the engagement characteristics and textual properties of the scraped tweets. The bottom left panel displays the log-transformed distribution of likes for the 199 tweets that received at least one like, revealing a strongly right-skewed engagement pattern where the majority of tweets received very few likes concentrated at log values below 2, while a small number of tweets achieved substantially higher engagement reaching log values above 6, corresponding to several hundred likes. This distribution is typical of social media discourse where a small proportion of highly viral posts dominates total engagement while the majority of content receives minimal attention. The bottom right panel shows the distribution of tweet text lengths in characters, with a median of 233 characters marked by the red dashed line. The distribution is notably bimodal, with one concentration around 100 to 150 characters representing shorter posts and a more prominent cluster between 250 to 300 characters approaching the platform character limit, suggesting that discourse about poverty statistics on X tends toward either brief reactive comments or more elaborated posts that maximize the available character allowance. The median text length of 233 characters indicates that on average the scraped tweets contain substantive textual content suitable for sentiment classification using the IndoRoBERTa model applied in the subsequent analysis.

B. Dimensionality Reduction and GMM Clustering

PCA decomposition of the 10 standardized socioeconomic indicators reveals that the first principal component alone accounts for 51.5% of total variance, with PC1 through PC3 cumulatively explaining 81.4%, consistent with the elbow point detected at PC3. UMAP stability analysis across n_components = 2 to 10 with five random seeds identifies n = 3 as the most stable configuration (mean Silhouette = 0.500, std = 0.024). GMM clustering is subsequently evaluated across k = 2 to 10 on both PCA and UMAP embeddings. As summarized in Table 2, UMAP+GMM consistently outperforms PCA+GMM across all cluster evaluation metrics at every value of k. The optimal configuration at k = 2 yields UMAP+GMM Silhouette Score = 0.822, Davies-Bouldin Index = 0.159, and Calinski-Harabasz Score = 593.561, compared to PCA+GMM values of 0.580, 0.452, and 117.912 respectively, confirming UMAP's superior capacity to preserve nonlinear socioeconomic structure for probabilistic provincial clustering.

Table 2. GMM Clustering Evaluation Metrics: PCA vs UMAP (k = 2 to 10)

K	Sil-PCA	DB-PCA	CH-PCA	Sil-UMAP	DB-UMAP	CH-UMAP
2	0.580	0.452	117.912	0.749	0.219	282.478
3	0.344	1.128	91.720	0.567	0.607	382.787
4	0.365	0.939	111.716	0.492	0.642	336.585
5	0.310	1.160	94.080	0.568	0.538	575.032
6	0.351	1.190	97.155	0.567	0.496	649.674
7	0.374	0.834	119.696	0.610	0.477	992.815
8	0.416	0.777	124.13	0.564	0.547	951.134

			7			
9	0.474	0.710	154.73	0.608	0.520	1236.05
10	0.453	0.979	137.60 1	0.610	0.534	1499.02 3

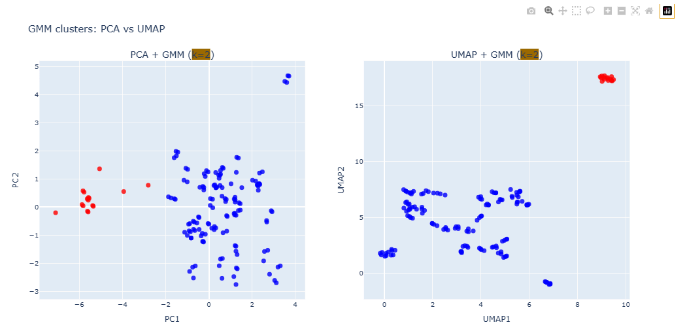

Figure 3. GMM Cluster Visualization: 2D Projections for PCA and UMAP

B. Ensemble Forecasting and SHAP Interpretability

Table 3 summarizes individual and ensemble model performance on the validation set. Random Forest achieves the lowest individual error (RMSE = 1.220, MAE = 0.573, MAPE = 3.86%), followed by XGBoost (RMSE = 1.527, MAE = 0.749, MAPE = 4.72%) and LightGBM (RMSE = 2.248, MAE = 1.393, MAPE = 12.55%). The weighted ensemble, with model weights of 0.427 (RF), 0.341 (XGBoost), and 0.232 (LightGBM) derived from inverse cross-validation RMSE, yields RMSE = 1.461, MAE = 0.659, and MAPE = 4.57%. While LightGBM's relatively weaker performance on this panel dataset reduces the ensemble's advantage over RF alone, the ensemble remains preferred for its robustness across provincial clusters. Selected 2025 poverty rate forecasts include Aceh (13.26%), Bali (4.11%), Banten (5.53%), Bengkulu (12.95%), and DI Yogyakarta (10.31%), reflecting the persistent regional disparity between eastern high-poverty and western low-poverty provinces.

Table 3. Ensemble Model Performance on Validation Set

Model	RMSE	MAE	MAPE (%)
RF	1.2202	0.5728	3.8582%
XGBoost	1.5268	0.7495	4.7195%
LightGBM	2.2476	1.3934	12.5472%
Ensemble Weighted	1.4609	0.6590	4.5698%

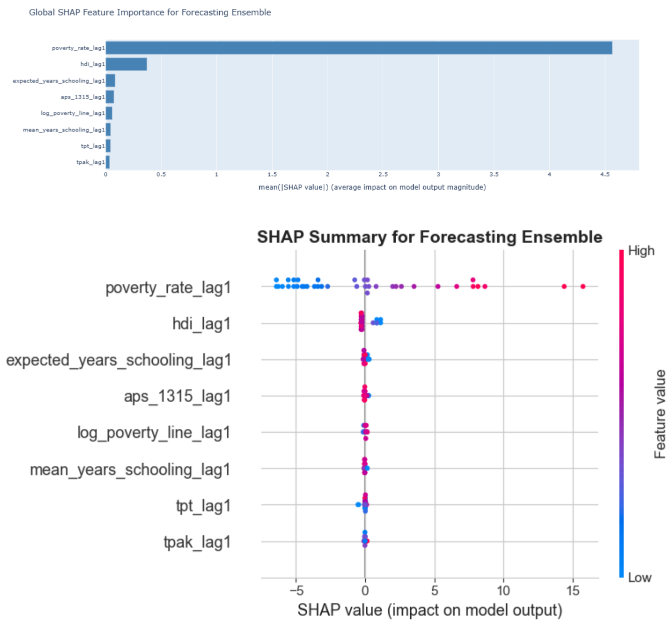

Figure 4. Global SHAP Feature Importance and Beeswarm Plot for Forecasting Ensemble

SHAP analysis reveals a strong path dependency structure in the ensemble predictions. The lagged poverty rate (poverty_rate_lag1) is the overwhelmingly dominant predictor with a mean absolute SHAP value of approximately 4.5, dwarfing all other features. HDI (hdi_lag1) emerges as the only other meaningfully contributing feature (~0.35), while remaining indicators - including TPT, TPAK, APS, and schooling variables - contribute negligibly. This pattern holds consistently across both the global validation set and the 2025–2026 forecast horizon, as well as across both GMM-derived provincial clusters (Figure 5), suggesting that regardless of a province's socioeconomic typology, its current poverty trajectory is most powerfully determined by its own prior poverty level and human development attainment.

Figure 5. SHAP Feature Importance by Provincial Cluster (2025–2026)

C. Spatial Autocorrelation

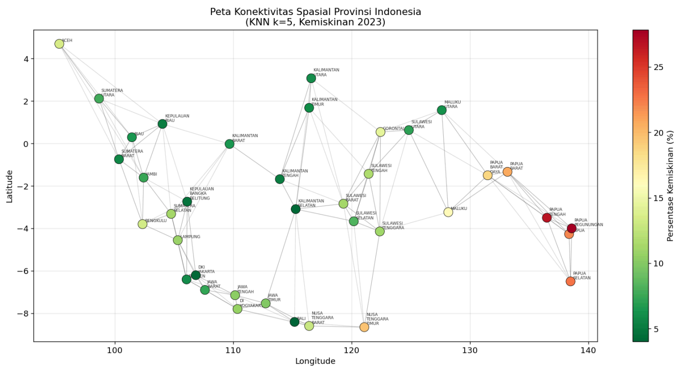

Figure 6. Spatial Connectivity Map of Indonesian Provinces Using K-Nearest Neighbors Weight Matrix (k=5, Poverty Rate 2023)

Figure 6 presents the spatial connectivity map of 38 Indonesian provinces constructed using a K-Nearest Neighbors (KNN) spatial weight matrix with k=5, based on 2023 poverty data. Each node represents a province, with node color encoded by poverty rate ranging from dark green (low poverty, approximately 5%) to dark red (high poverty, above 25%). The gray edges connecting nodes represent the five nearest spatial neighbors assigned to each province, resulting in a uniform average of 5.0 neighbors per province with a matrix sparsity of 13.16%, confirming a moderately sparse but well-connected spatial structure across the archipelago.

Several notable spatial patterns emerge from this connectivity map. Provinces in Java and Sumatra form tightly clustered subgraphs in the western portion of the map, reflecting their geographic proximity and consistently low poverty rates shown by the dominance of dark green nodes in those regions. In contrast, the eastern cluster comprising Papua Tengah, Papua Pegunungan, Papua, Papua Selatan, and Papua Barat Daya is characterized by predominantly red and orange nodes, visually confirming the concentration of high poverty in that region. The KNN connectivity structure further reveals that high-poverty Papua provinces are predominantly connected to one another, suggesting strong spatial neighborhood effects that will be formally quantified through Moran's I and LISA analysis in subsequent sections. Nusa Tenggara Timur, positioned as a bridge node between the central and eastern clusters, also exhibits an elevated poverty rate consistent with its ranking in the top ten provinces identified in Figure 2.

Table 4. Global Moran's I Results by Year (2021–2024)

Year	N	Moran's I	Expected I	Z-Score	p-value (sim)	Significant
2021	37	0.640870	−0.027778	7.44	0.001	Yes
2022	38	0.640874	−0.027027	7.54	0.001	Yes
2023	38	0.638448	−0.027027	7.51	0.001	Yes
2024	38	0.578220	−0.027027	6.83	0.001	Yes

Table 4 reports the Global Moran's I results for provincial poverty rates across all four observation years from 2021 to 2024. The Moran's I values are consistently high and positive across the entire period, registering 0.6409 in 2021, 0.6409 in 2022, 0.6384 in 2023, and 0.5782 in 2024, all substantially above the expected value under spatial randomness which hovered around -0.027 throughout the period. These values indicate strong positive spatial autocorrelation in provincial poverty distribution, meaning that

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provinces with high poverty rates tend to be geographically clustered together, and likewise provinces with low poverty rates cluster among themselves, rather than being randomly dispersed across the Indonesian archipelago. The magnitude of these values, consistently exceeding 0.57 across all years, places the observed spatial clustering well within the range typically characterized in the spatial econometrics literature as strong positive autocorrelation, underscoring that poverty in Indonesia is not a randomly distributed phenomenon but rather a geographically structured one with clear regional concentration patterns.

The statistical significance of these findings is robust across all years without exception. Z-scores ranged from 6.84 in 2024 to 7.54 in 2022, all far exceeding the conventional threshold of 1.96 for a 95% confidence level and even the more stringent threshold of 3.29 for a 99.9% confidence level. Both the normal approximation p-values, which approached zero at as low as 4.53×10^{-14} in 2022, and the simulation-based p-values derived from 999 permutation randomizations ($p = 0.001$ in all years) confirm that the observed spatial clustering pattern is extremely unlikely to arise by chance. This dual validation through both analytical and simulation-based inference strengthens the credibility of the spatial autocorrelation finding considerably, as the permutation-based test makes no distributional assumptions about the data and is particularly appropriate for the small sample of 38 provinces where asymptotic normality cannot be guaranteed.

A notable trend visible across the four-year observation window is the gradual decline in Moran's I from 0.6409 in 2021 and 2022 to 0.6384 in 2023 and further to 0.5782 in 2024, accompanied by a corresponding decrease in the Z-score from 7.54 in 2022 to 6.84 in 2024. While the spatial clustering of poverty remains strong and statistically significant throughout the entire observation period, this downward trajectory suggests a modest but discernible weakening of geographic poverty concentration over time, potentially reflecting incremental convergence between high-poverty eastern provinces and their neighbors as regional development programs and fiscal transfer policies gradually take effect. The most pronounced drop occurs between 2023 and 2024, where Moran's I fell by approximately 0.06 points, which may warrant further investigation in future studies to determine whether this represents a structural shift in spatial poverty dynamics or a temporary fluctuation. Nevertheless, the persistently high Moran's I values above 0.57 across all years confirm that spatial spillover effects remain a dominant structural feature of Indonesian provincial poverty, providing strong empirical justification for the LISA analysis conducted in the subsequent section.

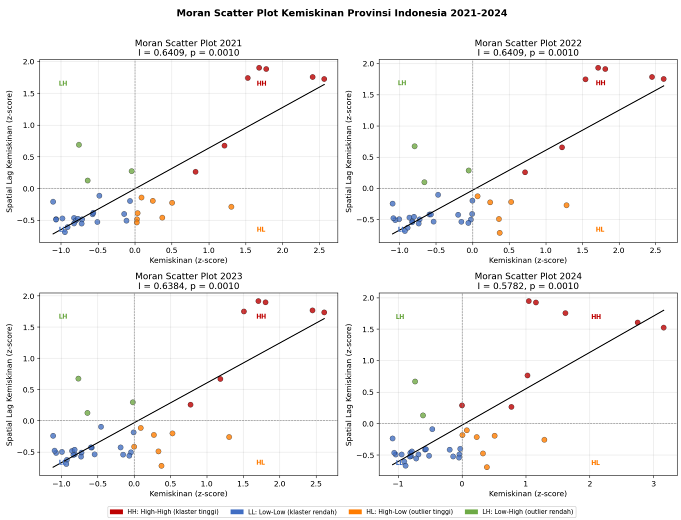

Figure 7. Moran Scatter Plot of Provincial Poverty Rates in Indonesia 2021-2024

Figure 7 presents the Moran scatter plots for provincial poverty rates across all four observation years from 2021 to 2024, where the horizontal axis represents the standardized poverty rate (z-score) of each province and the vertical axis represents the spatial lag, that is the weighted average poverty rate of each province's neighbors. The slope of the fitted regression line in each panel corresponds directly to the Global Moran's I statistic reported in Table 4, and the four quadrants of each plot classify provinces into their respective LISA categories: High-High (HH, red) indicating provinces with high poverty surrounded by high-poverty neighbors, Low-Low (LL, blue) indicating provinces with low poverty surrounded by low-poverty neighbors, High-Low (HL, orange) indicating high-poverty provinces surrounded by relatively low-poverty neighbors, and Low-High (LH, green) indicating low-poverty provinces surrounded by relatively high-poverty neighbors.

Across all four years, the scatter plots reveal a consistent and pronounced positive linear relationship between a province's own poverty rate and the spatial lag of its neighbors, as evidenced by the steep upward-sloping regression line in each panel. The High-High quadrant in the upper right is persistently occupied by a tight cluster of red points representing the Papua provinces, which exhibit both the highest own poverty rates and the highest neighbor poverty rates simultaneously, confirming their role as the core of the spatial poverty cluster in eastern Indonesia. The Low-Low quadrant in the lower left is densely populated by blue points representing Java, Bali, and several Sumatra provinces, reflecting the strong geographic concentration of low-poverty provinces in the western archipelago. The relative stability of point positions across the four panels indicates that provincial poverty typologies have remained structurally persistent throughout the 2021 to 2024 period with very limited mobility between quadrants.

The High-Low quadrant in the lower right, occupied by orange points, contains a smaller number of provinces that represent spatial anomalies, provinces with relatively elevated poverty rates despite being surrounded by neighbors with lower poverty rates. Similarly, the Low-High quadrant in the upper left contains green points representing provinces with comparatively lower poverty that are nevertheless neighbored by higher-poverty provinces, such as certain Sulawesi provinces adjacent to the eastern cluster. The gradual flattening of the regression slope from 2021 to 2024, consistent with the declining Moran's I values reported in Table 4, is also faintly visible across the four panels, particularly in the 2024 plot where the rightmost HH points appear slightly less vertically elevated compared to earlier years. Taken together, Figure 7 reinforces the conclusion that spatial clustering of poverty in Indonesia is not only statistically significant but also spatially structured in a highly stable and geographically coherent manner across the entire observation period.

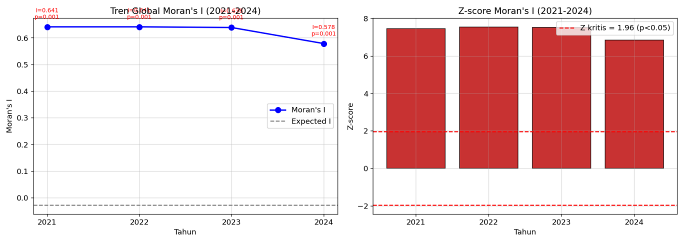

Figure 8. Global Moran's I Trend and Z-score of Provincial Poverty Rates in Indonesia (2021-2024)

Figure 8 (left) illustrates the temporal trend of Global Moran's I values from 2021 to 2024, plotted against the expected value under spatial randomness represented by the gray dashed line at approximately -0.027. The blue line connecting the four annual Moran's I estimates remains nearly flat between 2021 and 2023, with values of 0.641, 0.641, and 0.638 respectively, before declining more sharply to 0.578 in 2024. The wide and consistent gap between the observed Moran's I line and the expected value baseline across all four years visually confirms the magnitude of spatial autocorrelation, as even the lowest recorded value in 2024 remains more than 0.60 units above what would be expected under a spatially random distribution. All four

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data points are annotated with their corresponding Moran's I value and simulation-based p-value of 0.001, providing an at-a-glance confirmation that the spatial clustering finding is statistically significant in every observation year without exception.

Figure 8 (right) complements the trend plot by displaying the Z-scores of the Moran's I statistic for each year as red bar charts, with the critical Z threshold of 1.96 ($p < 0.05$) marked by the red dashed horizontal lines at +1.96 and -1.96. The Z-scores for all four years, reaching approximately 7.44 in 2021, 7.54 in 2022, 7.52 in 2023, and 6.84 in 2024, tower far above the upper critical threshold, with all bars extending well beyond the 6.0 mark. This visualization makes immediately apparent how far the observed spatial clustering departs from the null hypothesis of spatial randomness, as even the lowest Z-score in 2024 is more than three times the critical value required for significance at the 95% confidence level. The slight reduction in bar height from 2022 to 2024 mirrors the declining trend observed in the left panel, reinforcing the interpretation that while spatial poverty clustering in Indonesia is weakening marginally over time, it remains an overwhelmingly dominant and statistically unambiguous structural feature of the provincial poverty landscape throughout the entire 2021 to 2024 observation period.

Figure 9 presents the LISA maps of provincial poverty rates across all four observation years, where only statistically significant provinces ($p < 0.05$) are rendered in color while non-significant provinces appear as gray points. In 2021, 6 out of 37 provinces reached significance, comprising 5 High-High provinces and 1 Low-Low province. The five HH provinces, namely Papua Pegunungan ($I = 4.30$), Papua Tengah ($I = 4.11$), Papua Selatan ($I = 3.26$), Papua ($I = 3.11$), and Papua Barat ($I = 2.59$), all recorded extremely high local Moran's I values with $p = 0.001$, confirming their status as the core of the most structurally entrenched poverty hotspot in the Indonesian archipelago. Kalimantan Tengah emerged as the sole Low-Low province in 2021 with $I = 0.635$ ($p = 0.026$), identifying it as part of a statistically significant coldspot of low poverty in the Kalimantan region.

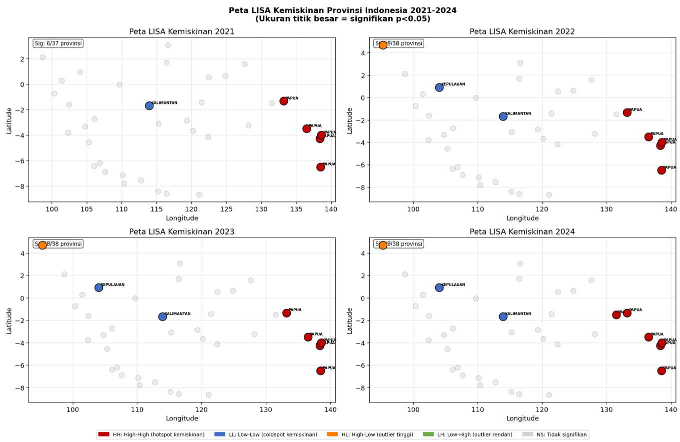

Figure 9. LISA Cluster Map of Provincial Poverty Rates (2021–2024)

From 2022 onward, the number of significant provinces expanded to 8 out of 38, with the HH cluster in Papua remaining stable while two new categories emerged. Kepulauan Riau joined Kalimantan Tengah in the Low-Low quadrant in both 2022 and 2023, reflecting a growing and statistically coherent low-poverty cluster in the western archipelago. Aceh appeared as a High-Low spatial anomaly in 2022 and persisted through 2023 and 2024 with consistently negative local Moran's I values around -0.26 and p-values between 0.022 and 0.028, indicating that Aceh maintains a relatively elevated poverty rate despite being surrounded by lower-poverty Sumatra neighbors, classifying it as a spatial outlier rather than part of a broader poverty cluster. The composition of the HH Papua cluster remained unchanged across 2022 and 2023, with all five provinces retaining $p = 0.001$ and local I values increasing slightly, suggesting that the spatial concentration of poverty in Papua intensified rather than weakened during this period despite the declining global Moran's I.

The 2024 LISA map marks the most notable structural shift across the observation period, with significant provinces increasing to 9 out of 38. Papua Barat Daya entered the High-High quadrant for the first time with I = 0.761 (p = 0.046), expanding the Papua poverty hotspot from five to six provinces and indicating that the spatial poverty cluster in eastern Indonesia has grown in geographic extent even as the global Moran's I value declined slightly. Meanwhile, the local I values of Papua Pegunungan and Papua Tengah continued to rise, reaching 4.680 and 4.294 respectively in 2024, the highest values recorded across the entire observation period, reinforcing that the core of the Papua poverty cluster is intensifying at the provincial level. The simultaneous expansion of the HH cluster and slight weakening of the global Moran's I in 2024 is not contradictory. It reflects a pattern where the eastern poverty hotspot is becoming more internally concentrated while other provinces across the archipelago may be experiencing marginal convergence, a finding that carries significant implications for spatially targeted poverty alleviation policies in eastern Indonesia.

Figure 10 presents the LISA quadrant consistency matrix for all provinces that were statistically significant at least once across the 2021 to 2024 observation period. Each row represents a province and each column represents a year, with cell colors indicating the LISA quadrant classification: red for High-High, blue for Low-Low, orange for High-Low, and gray for Not Significant. The visualization reveals a striking degree of temporal stability in quadrant membership, with the majority of provinces retaining their assigned classification consistently across all four years, providing strong evidence that spatial poverty typologies in Indonesia are structurally persistent rather than transient.

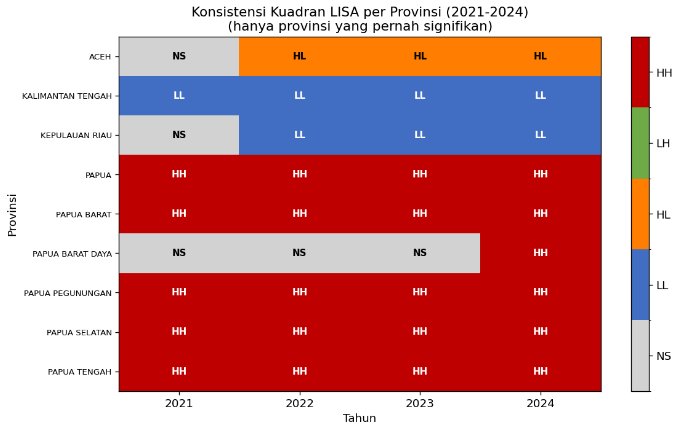

Figure 10. LISA Quadrant Consistency per Province (2021-2024)

The High-High cluster demonstrates the most absolute consistency across the observation period. Papua, Papua Barat, Papua Pegunungan, Papua Selatan, and Papua Tengah all maintained HH classification without a single year of deviation from 2021 through 2024, confirming that these five provinces constitute the most entrenched and spatially coherent poverty hotspot in Indonesia. Their uninterrupted HH status across all four years, each with p = 0.001, indicates that membership in this poverty cluster is not sensitive to annual fluctuations in poverty rates but reflects a deep structural spatial dependency that persists regardless of short-term changes. Papua Barat Daya presents an interesting transitional case, classified as Not Significant in 2021, 2022, and 2023 before shifting to HH in 2024, suggesting a gradual absorption into the core Papua poverty cluster as its spatial association with neighboring high-poverty provinces strengthened over time.

The Low-Low and High-Low provinces exhibit similarly consistent patterns, though with slight variations in their year of entry into significance. Kalimantan Tengah maintained LL classification across all four years including 2021, making it the most temporally stable coldspot in the dataset. Kepulauan Riau entered the LL quadrant from 2022 onward and retained that classification through 2024, indicating a consolidating low-poverty spatial cluster in the Riau archipelago region. Aceh transitioned from Not Significant in 2021 to a persistent HL classification from 2022 through 2024, identifying it as a stable spatial outlier where elevated provincial poverty exists within a neighborhood of relatively lower-poverty provinces. The overall pattern across Figure 10 reinforces the conclusion that spatial poverty structures in Indonesia are highly path-dependent, with provinces rarely crossing quadrant boundaries once they achieve statistical significance, and that targeted regional interventions must account for the self-reinforcing geographic nature of both poverty hotspots and coldspots across the archipelago.

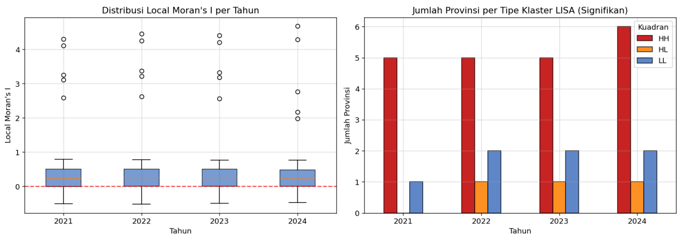

Figure 11. Distribution of Local Moran's I per Year and Number of Provinces per Significant LISA Cluster Type (2021-2024)

Figure 11 (left) presents the annual distribution of Local Moran's I values across all provinces from 2021 to 2024 through box plots, with the red dashed line at zero serving as the reference threshold separating positive spatial association from negative. The bulk of provincial distributions remains tightly concentrated just above zero across all four years, with median values consistently around 0.25 to 0.30, indicating that the majority of Indonesian provinces exhibit modest positive local spatial association with their neighbors. However, the presence of pronounced upper outliers in every year, reaching as high as 4.10 to 4.68, reflects the extreme local Moran's I values generated by the Papua cluster provinces identified in the LISA analysis. The overall interquartile range remains narrow and stable across the four years, suggesting that for most provinces the degree of local spatial association with neighbors has not changed meaningfully over the observation period, and that the extreme outliers in the upper tail are driven almost entirely by the persistent Papua poverty hotspot rather than by a general intensification of spatial clustering across the archipelago.

Figure 11 (right) provides a complementary count-based perspective by displaying the number of statistically significant provinces per LISA quadrant type for each year. The HH cluster, shown in red, consistently constitutes the largest category of significant provinces throughout the observation period, comprising 5 provinces in 2021, 2022, and 2023 before expanding to 6 provinces in 2024 with the addition of Papua Barat Daya. The LL cluster, shown in blue, grew from 1 province in 2021 to 2 provinces from 2022 onward as Kepulauan Riau joined Kalimantan Tengah, and remained stable at 2 through 2024. The HL cluster, shown in orange, appeared from 2022 onward with a consistent count of 1 province representing Aceh across all three subsequent years. Taken together, the right panel confirms that the total number of spatially significant provinces has grown incrementally from 6 in 2021 to 9 in 2024, driven primarily by the expansion of the HH cluster, and that the spatial structure of Indonesian provincial poverty is characterized by a dominant and growing poverty hotspot in the east, a modest but stable coldspot in the west, and a single persistent spatial outlier in northern Sumatra throughout the entire observation period.

D. Fixed Effect Panel Regression

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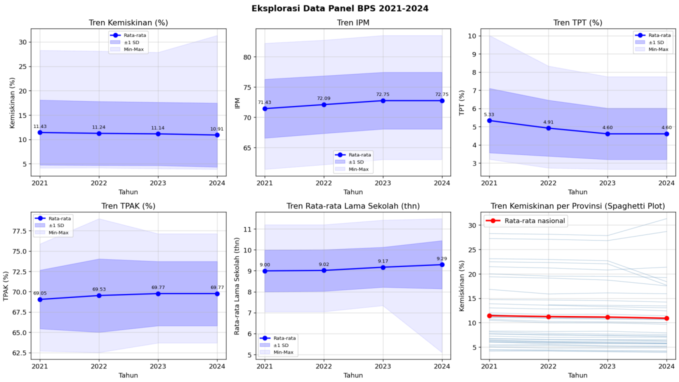

Figure 12. BPS Panel Data Exploration 2021-2024

Figure 12 presents the temporal trends of five key socioeconomic indicators across 38 Indonesian provinces from 2021 to 2024, alongside a spaghetti plot of provincial poverty trajectories. The poverty rate trend (top left) shows a gradual national average decline from 11.43% in 2021 to 10.91% in 2024, a reduction of approximately 0.52 percentage points over the four-year period. While this downward trend is encouraging, the wide min-max band extending from approximately 4% to above 30% throughout all years underscores the enormous disparity between the lowest and highest poverty provinces, and the relatively narrow one standard deviation band around the mean suggests that the bulk of provinces cluster around the 10% range while a small number of extreme outlier provinces in eastern Indonesia drive the upper bound of the distribution considerably higher.

The Human Development Index trend (top center) shows a consistent upward trajectory from a national average of 71.43 in 2021 to 72.75 in 2024, reflecting steady improvement in the composite measure of health, education, and purchasing power across provinces. The Open Unemployment Rate (top right) declined from 5.33% in 2021 to 4.60% in 2024, with the sharpest drop occurring between 2021 and 2022 as post-pandemic labor market recovery took hold. The Labor Force Participation Rate (bottom left) exhibited a modest but consistent increase from 69.05% in 2021 to 69.77% in 2023 before plateauing at 69.77% in 2024, indicating gradual expansion of productive labor force engagement across provinces. The Mean Years of Schooling (bottom center) likewise trended upward from 9.00 years in 2021 to 9.29 years in 2024, reflecting incremental improvements in educational attainment at the provincial level. Across all five indicators, the direction of change is consistent with conditions theoretically conducive to poverty reduction, providing preliminary descriptive support for the causal relationships to be formally tested in the Fixed Effect Panel Regression.

The spaghetti plot (bottom right) offers the most direct visualization of within-province poverty dynamics over time, with each thin blue line representing an individual province trajectory and the thick red line representing the national average. The majority of provincial lines run nearly flat and parallel to the national average in the lower portion of the chart, indicating that most provinces experienced minimal change in poverty rates over the four years. However, a small number of lines in the upper portion of the chart display more pronounced downward slopes, corresponding to the high-poverty Papua and eastern Indonesia provinces that recorded larger absolute reductions from their elevated baselines. The tight clustering of most provincial trajectories around and below the national average, combined with the few steeply positioned lines in the upper range, visually reinforces the persistent structural inequality between western and eastern Indonesian provinces that motivates the use of Two-Way Fixed Effect estimation to control for time-invariant province-level heterogeneity in the regression analysis that follows.

Prior to estimating the Fixed Effect Panel Regression model, a variance decomposition analysis was conducted to assess the proportion of total variation in each variable attributable to between-province differences versus within-province changes over time. This

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decomposition is critical for justifying the use of Fixed Effect estimation, as the within-province variation is the only source of identification in a Fixed Effect model after province-level fixed effects are absorbed. The results reveal that for the dependent variable, poverty rate (y_kemiskinan), the between-province standard deviation of 6.512 is nearly identical to the total standard deviation of 6.491, while the within-province standard deviation is only 0.573, yielding a within-to-total ratio of just 8.83%. This indicates that the vast majority of variation in poverty rates across the dataset is driven by permanent structural differences between provinces rather than by year-to-year changes within individual provinces, a pattern that strongly motivates the inclusion of province fixed effects to control for these time-invariant characteristics.

A similar pattern is observed across all explanatory variables. The Human Development Index exhibits a within-to-total ratio of 11.27%, the Labor Force Participation Rate 20.14%, the Open Unemployment Rate 24.85%, and the log of the Poverty Line 21.55%, all indicating that between-province heterogeneity accounts for the dominant share of variation in each variable. The School Participation Rate for the 13 to 15 age group records the lowest within variation ratio at 5.01%, suggesting that differences in school participation between provinces are almost entirely structural and time-invariant over the observation period. In contrast, Mean Years of Schooling and Expected Years of Schooling exhibit the highest within variation ratios at 33.57% and 34.26% respectively, indicating that these education attainment variables experienced relatively more meaningful within-province change over the four years compared to other indicators. While the within variation ratios are universally modest given the short four-year panel, the existence of non-trivial within-province variation across all variables confirms that Fixed Effect estimation remains feasible and that the model retains sufficient temporal variation to identify the causal effects of interest after controlling for permanent provincial heterogeneity.

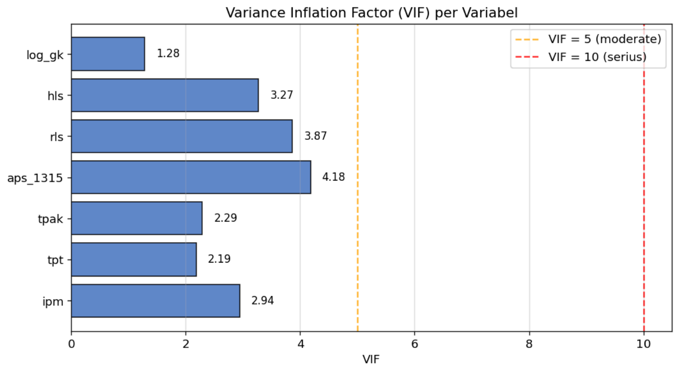

Figure 13. Variance Inflation Factor (VIF) per Variable

Figure 13 presents the Variance Inflation Factor (VIF) values for all seven independent variables included in the Fixed Effect Panel Regression model. VIF measures the degree to which each variable's variance is inflated due to linear correlation with other predictors, with values above 5 conventionally indicating moderate multicollinearity and values above 10 indicating serious multicollinearity that may destabilize coefficient estimates. All seven variables in the model fall well below both thresholds, with VIF values ranging from a minimum of 1.28 for log_gk to a maximum of 4.18 for aps_1315, confirming that multicollinearity does not pose a threat to the reliability of the regression estimates in this study.

Among the variables, aps_1315 records the highest VIF at 4.18, followed by rls at 3.87 and hls at 3.27, which is expected given that school participation rate, mean years of schooling, and expected years of schooling are all education-related indicators that naturally share a degree of correlation. IPM registers a VIF of 2.94, reflecting its composite nature as an index that incorporates education and health components which overlap conceptually with other predictors in the model. The labor

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market variables tpak and tpt record the two lowest VIF values among the non-poverty-line variables at 2.29 and 2.19 respectively, indicating minimal linear dependence with other regressors. Log_gk exhibits the lowest VIF of all at 1.28, suggesting that the log-transformed poverty line is largely orthogonal to the remaining predictors. The absence of any variable exceeding the moderate threshold of 5 provides assurance that all seven independent variables can be retained in the model without risk of multicollinearity-induced coefficient instability or inflated standard errors in the Fixed Effect estimation that follows.

The estimation procedure followed a sequential model comparison approach beginning with Pooled OLS as a baseline, followed by entity-only Fixed Effects, Two-Way Fixed Effects incorporating both entity and time effects, and Random Effects, with the optimal specification determined through formal statistical testing. The Pooled OLS baseline model yielded an overall R-squared of 0.739 and identified ipm, tpt, tpak, and hls as significant predictors, however its negative within R-squared of -2.155 immediately signals that the model fails to account for within-province variation and is heavily distorted by between-province heterogeneity, making it an unreliable basis for causal inference. The F-test for poolability from the Fixed Effects estimation returned an F-statistic of 859.55 (p = 0.000), firmly rejecting the null hypothesis that Pooled OLS is sufficient and confirming that province-level fixed effects are statistically necessary.

The Hausman Test formally adjudicated between Fixed Effects and Random Effects specifications, yielding an H-statistic of 31.583 with 7 degrees of freedom and p-value of 0.000. This result decisively rejects the null hypothesis of no correlation between individual effects and the regressors, confirming that Random Effects estimates would be inconsistent and that Fixed Effects is the appropriate estimator for this dataset. This conclusion is further reinforced by the Random Effects within R-squared of 0.886, which closely approximates the Fixed Effects within R-squared of 0.889, suggesting that while Random Effects performs similarly in terms of fit, it cannot be trusted for inference given the confirmed endogeneity of the individual effects. The Two-Way Fixed Effects model, which additionally controls for time fixed effects, achieves the highest overall R-squared of 0.916 and an F-statistic for poolability of 1176.3 (p = 0.000), and is therefore adopted as the preferred final specification for all subsequent interpretation.

The Two-Way Fixed Effects estimation results reveal that rls (Mean Years of Schooling) and log_gk (log Poverty Line) are the two statistically significant drivers of within-province poverty dynamics after controlling for all time-invariant provincial characteristics and common time shocks. The coefficient on rls is -1.225 (p = 0.000), indicating that a one-year increase in mean years of schooling within a province is associated with a reduction of 1.225 percentage points in the poverty rate, holding all else constant. This is the strongest and most precisely estimated effect in the model, with a t-statistic of -17.375 and a 95% confidence interval of -1.365 to -1.085, confirming that improvements in educational attainment are the most robust within-province determinant of poverty reduction across Indonesian provinces from 2021 to 2024. The coefficient on log_gk is 8.820 (p = 0.004), indicating that a one-unit increase in the log poverty line is associated with an 8.820 percentage point increase in measured poverty, which reflects the mechanical relationship between the poverty threshold and the measured poverty headcount ratio rather than a causal deterioration in welfare. The coefficient on tpt

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is -0.163 (p = 0.001), indicating that a one percentage point reduction in the open unemployment rate within a province is associated with a 0.163 percentage point reduction in poverty, though this effect is modest in magnitude compared to rls. Variables including ipm, tpak, aps_1315, and hls did not achieve statistical significance in the Two-Way Fixed Effects specification, suggesting that after absorbing province and time fixed effects, their within-province variation over the short four-year window is insufficient to independently identify their effects on poverty beyond what is already captured by rls and log_gk.

E. Sentiment Analysis

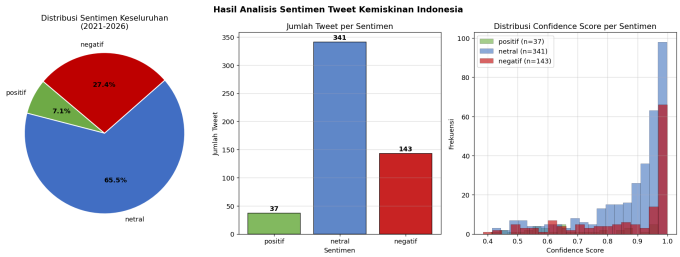

Figure 14. Sentiment Analysis Results of Indonesian Poverty-Related Tweets (2021-2026)

Figure 14 presents the results of the IndoRoBERTa-based sentiment classification applied to all 521 poverty-related tweets scraped from X covering 2021 to 2026. The pie chart (left) and bar chart (center) jointly reveal that neutral sentiment constitutes the dominant category, comprising 341 tweets or 65.5% of the total corpus, followed by negative sentiment at 143 tweets or 27.4%, and positive sentiment as the smallest category at 37 tweets or 7.1%. The overwhelming dominance of neutral sentiment is consistent with the nature of the scraped content, as tweets containing the keyword "persentase penduduk miskin" tend to consist largely of factual reporting, statistical citation, news dissemination, and academic or policy-oriented commentary rather than emotionally charged personal expressions. This reflects the relatively technical and data-driven character of poverty discourse on Indonesian social media when framed around official statistical terminology, in contrast to more colloquial poverty-related keywords that would likely elicit stronger emotional responses.

The negative sentiment category at 27.4% represents a substantial minority of the corpus and captures public expressions of concern, criticism, and dissatisfaction related to poverty conditions in Indonesia. The presence of 143 negatively classified tweets across the 2021 to 2026 period suggests that a meaningful portion of public discourse surrounding official poverty statistics reflects frustration with the pace of poverty reduction, skepticism toward government policy, or direct commentary on the hardship experienced by affected communities. The positive sentiment category, comprising only 37 tweets at 7.1%, represents the smallest share and likely captures expressions of optimism regarding poverty reduction progress, acknowledgment of policy achievements, or commentary on improvements in specific regions or indicators. The stark asymmetry between negative and positive sentiment, with negative tweets outnumbering positive ones by nearly four to one, indicates that public discourse on poverty statistics in Indonesia leans considerably more toward critical and concerned sentiment than toward optimistic or affirmative framing throughout the entire observation period.

The confidence score distribution (right panel) provides strong validation of the classification quality produced by the IndoRoBERTa model. The overall mean confidence score across all 521 tweets is 0.856 with a standard deviation of 0.158, and 95.8% of all classifications achieved a confidence score above 0.5, indicating that the model assigned its labels with high certainty in the vast majority of cases. The distribution is heavily right-skewed, with the median confidence of 0.929 and the 75th percentile at

0.975, reflecting that most predictions clustered near the maximum confidence of 0.999. Disaggregating by sentiment class, neutral tweets achieved the highest mean confidence at 0.871, followed closely by negative tweets at 0.860, while positive tweets recorded the lowest mean confidence at 0.697. The lower confidence for positive classifications is noteworthy and suggests that the model found positive sentiment in this corpus more ambiguous to identify, which may reflect the relative scarcity of clearly positive discourse in poverty-related content and the potential overlap between mildly positive and neutral tones in factual reporting contexts. The 22 tweets with confidence below 0.5, representing only 4.2% of the corpus, constitute borderline cases where the model was genuinely uncertain, and their small proportion relative to the full dataset indicates that the overall sentiment classification results are robust and reliable for temporal trend analysis in the subsequent section.

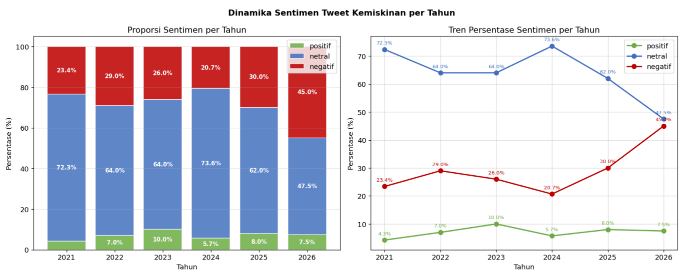

Figure 15. Sentiment Dynamics of Poverty-Related Tweets per Year (2021-2026)

Figure 15 presents the annual dynamics of sentiment distribution across the 521 poverty-related tweets from 2021 to 2026, displayed as both proportional stacked bar charts (left) and a multi-line trend plot (right). Neutral sentiment remained the dominant category throughout the observation period, though its share exhibited notable fluctuations rather than a stable plateau. Starting at 72.3% in 2021, neutral sentiment declined to 64.0% in both 2022 and 2023, recovered sharply to 73.6% in 2024, then dropped again to 62.0% in 2025 before falling dramatically to 47.5% in 2026. This oscillating pattern suggests that the proportion of factual and informational discourse about poverty statistics is sensitive to the broader media and policy context of each year, with certain years generating more emotionally charged commentary that displaces neutral framing. The sharp decline in neutral sentiment in 2026 to below 50% is particularly noteworthy, though it must be interpreted with caution given that the 2026 data covers only a partial year with 40 tweets, making the 2026 proportions more susceptible to sampling variability than the more complete annual observations from 2021 to 2025.

Negative sentiment displayed a more volatile trajectory that broadly reflects periods of heightened public concern about poverty conditions in Indonesia. Beginning at 23.4% in 2021, negative sentiment rose to its first local peak of 29.0% in 2022 before moderating to 26.0% in 2023 and reaching its lowest point of 20.7% in 2024, the year in which neutral sentiment simultaneously peaked at 73.6%. This 2024 trough in negative sentiment coincides with the BPS data showing the lowest recorded national average poverty rate of 10.91% in the observation period, suggesting a plausible alignment between improving official poverty statistics and a temporary reduction in critical public discourse. However, negative sentiment then rebounded sharply to 30.0% in 2025 and surged to 45.0% in 2026, with the 2026 figure representing the highest negative proportion recorded across the entire series and approaching parity with neutral sentiment for the first time. Positive sentiment remained consistently low throughout all years, fluctuating narrowly between 4.3% in 2021 and 10.0% in 2023, never exceeding 10% in any complete annual observation, and averaging approximately 7% across the full period. The persistently marginal share of positive sentiment across all six years

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reinforces the overall characterization of Indonesian public discourse about poverty statistics as predominantly informational with a meaningful undercurrent of critical and negative sentiment that intensified toward the end of the observation window.

Chapter 4 Overall Discussion

The results of this study collectively demonstrate that provincial poverty in Indonesia is not a randomly distributed or independently determined phenomenon, but rather a geographically structured, temporally persistent, and multidimensionally driven condition that requires an integrated analytical framework to fully characterize. Across all five analytical modules, a consistent narrative emerges: poverty is concentrated in eastern Indonesia, spatially self-reinforcing, resistant to short-term change, and primarily driven by human capital accumulation at the within-province level. The UMAP and PCA comparison for GMM clustering established that UMAP consistently outperforms PCA in preserving the nonlinear socioeconomic structure of provincial data, with the optimal k=2 UMAP+GMM configuration yielding a Silhouette Score of 0.822 compared to 0.580 for PCA+GMM, confirming that dimensionality reduction choice is not a neutral methodological decision but one that materially affects the quality of resulting provincial typologies. The two GMM clusters broadly align with a western low-poverty cluster comprising Java, Bali, and most of Sumatra, and an eastern high-poverty cluster dominated by Papua and eastern Indonesia provinces, a bifurcation that recurs consistently across all subsequent modules and constitutes the structural backbone of the entire analytical system.

The spatial autocorrelation analysis provides the most direct evidence of geographic poverty lock-in. Global Moran's I values exceeding 0.57 across all four observation years, each with simulation-based p-values of 0.001, establish that provincial poverty clustering in Indonesia is among the most statistically robust spatial patterns observable in the national socioeconomic data. The LISA analysis further identifies a core of five to six Papua provinces as a persistent High-High hotspot with local Moran's I values reaching as high as 4.68 in 2024, values that dwarf all other provinces and indicate extreme spatial concentration of poverty that intensified rather than diminished over the observation period at the local level even as the global Moran's I declined marginally. This simultaneous global weakening and local intensification is a theoretically important finding: it suggests that while some degree of spatial convergence may be occurring at the periphery of the poverty distribution, the core hotspot provinces are becoming more internally homogeneous and spatially cohesive, implying that general national poverty reduction policies are unlikely to penetrate the Papua cluster without geographically targeted interventions that explicitly account for cross-provincial spillover effects.

The Fixed Effect Panel Regression results provide the causal complement to the spatial findings. After controlling for all time-invariant provincial characteristics and common macroeconomic shocks through Two-Way Fixed Effects, only mean years of schooling (rls) and the log poverty line (log_gk) emerge as statistically significant within-province drivers of poverty change. The coefficient of -1.225 on rls is the most precisely estimated effect in the model, with a t-statistic of -17.375 that leaves no statistical ambiguity: within Indonesian provinces, improvements in educational attainment measured by mean years of schooling are the single most powerful lever for poverty reduction observable in the 2021 to 2024 panel. This finding is consistent with and reinforces the SHAP analysis from the ensemble forecasting module, which identified

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hdi_lag1 as the only meaningful non-autoregressive predictor of poverty trajectories, with HDI incorporating an educational component that aligns directionally with the rls finding. The insignificance of ipm, tpak, aps_1315, and hls in the TWFE specification does not imply that these variables are unimportant in the cross-sectional sense, but rather that their within-province variation over the short four-year window is insufficient to identify independent causal effects after province and time fixed effects are absorbed, a limitation inherent to the brevity of the panel rather than to the variables themselves.

The sentiment analysis module, while operating on a separate data stream without integration into the tabular modeling pipeline, contributes a qualitatively important layer to the overall picture. The dominance of neutral sentiment at 65.5% across 521 tweets reflects the technical and statistical framing of public discourse around the keyword "persentase penduduk miskin," but the persistent undercurrent of negative sentiment at 27.4% across the full period, rising sharply to 45.0% in 2026, signals that public concern about poverty conditions in Indonesia is real, growing, and not fully captured by official statistics that show only gradual aggregate improvement. The notable alignment between the 2024 trough in negative sentiment (20.7%) and the lowest recorded national average poverty rate of 10.91% in the BPS data suggests a plausible responsiveness of public sentiment to official poverty statistics, lending partial cross-modal validation to the quantitative findings. Taken holistically, the five analytical modules of GeoSosEkon converge on a unified policy narrative: poverty in Indonesia is structurally geographic, educationally determined at the within-province level, spatially locked in the east, and increasingly perceived negatively by the public, all of which collectively argue for sustained, geographically differentiated, and education-centered poverty alleviation strategies that go well beyond aggregate national targeting.

Chapter 5 Conclusion and Recommendations

A. Conclusion

This study presents GeoSosEkon, a multimodal integrated spatial-temporal analytics system for modeling provincial poverty dynamics in Indonesia using BPS socioeconomic data from 2021 to 2024 and poverty-related tweets from 2021 to 2026. Across five parallel analytical modules, the system consistently identifies a structurally bifurcated poverty landscape in which a western cluster of low-poverty provinces and an eastern cluster dominated by Papua and surrounding provinces exhibit fundamentally different socioeconomic trajectories. The UMAP+GMM clustering, spatial autocorrelation analysis, and Fixed Effect Panel Regression all independently converge on the same geographic divide, confirming that provincial poverty clustering in Indonesia is not an artifact of any single analytical method but a robust empirical regularity that persists across machine learning, spatial statistics, and econometric frameworks simultaneously. The Moran's I values consistently above 0.57 with permutation-based p-values of 0.001 across all four years, combined with the uninterrupted High-High LISA classification of five core Papua provinces throughout 2021 to 2024, provide the strongest quantitative evidence that geographic spillover effects are a defining structural feature of Indonesian provincial poverty.

The causal inference component of the system establishes that mean years of schooling is the most robust within-province determinant of poverty reduction, with a TWFE coefficient of -1.225 ($p = 0.000$) that survived the most stringent econometric controls available for this panel structure. The ensemble forecasting module, with Random Forest achieving the lowest individual RMSE of 1.220 and the weighted ensemble yielding MAPE of 4.57%, demonstrates that reliable short-horizon poverty projections are achievable from the available socioeconomic indicators, with SHAP analysis revealing that prior poverty level and human development attainment jointly account for the overwhelming share of predictive power. The IndoRoBERTa sentiment analysis adds a qualitative temporal dimension, revealing that public discourse on poverty statistics in Indonesia is predominantly neutral but carries a significant and growing negative undercurrent that reached its highest recorded proportion of 45.0% in 2026, suggesting that public perception of poverty conditions does not fully track the gradual improvement reflected in official aggregate statistics.

B. Recommendations

Based on the findings of this study, several policy and research recommendations can be drawn. The persistent High-High spatial clustering of poverty in eastern Indonesia, particularly across Papua Pegunungan, Papua Tengah, Papua Selatan, Papua Barat, Papua, and Papua Barat Daya, calls for geographically differentiated poverty alleviation strategies that treat the eastern archipelago as an integrated regional system rather than isolated provincial cases, with fiscal transfers and social protection programs designed to leverage cross-provincial spillover effects. The TWFE coefficient of -1.225 on mean years of schooling, corroborated by SHAP-identified HDI importance in the forecasting module, identifies educational attainment as the most actionable within-province lever for poverty reduction, suggesting that programs extending school retention and progression beyond basic education should be prioritized, particularly in provinces with below-average schooling duration. Future studies are encouraged to extend the panel to a longer time series to increase within-province variation, incorporate province-level geolocation into the Twitter scraping pipeline to enable spatial sentiment mapping, and apply the GeoSosEkon framework at the regency and city level to capture finer-grained poverty dynamics that remain obscured at the provincial scale of the present analysis.

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Appendix

Drive Documentation

Github

Dashboard

Member of Contributions

Name	Contribution
Rizki Piji Fathoni	Ensemble Forecasting (Random Forest, XGBoost, LightGBM), SHAP Adaptive Interpretability, and Interactive Web Dashboard Development
Muhammad Raffi Fahrezi	Dimensionality Reduction (UMAP and PCA Comparison), GMM Clustering, and Project Presentation Initialization
Nazril Ravi Pratma	Manual BPS Data Collection, Twitter Data Scraping, Fixed Effect Panel Data Regression, Spatial Autocorrelation Analysis (Moran's I and LISA), and Sentiment Analysis using IndoRoBERTa